Computer Networks 2 – Exam Solutions March 2025

Maximum points: 37p
Preliminary grading: Pass (3): 19p, (4): 26p, (5): 33p

ANONYMOUS NUMBER:

All answers must be in English. Start a new page for each problem.

Read the questions carefully and answer what is asked for! Your task is to convince the teachers who are grading the exam that you have a certain level of knowledge. The best way to do this are concise, comprehensible and clear answers to the questions asked – and nothing else.
In some of the problems, you may not be given all the details needed to answer the problem. In such cases, you need to make realistic assumptions that are explicitly stated and motivated. Omitting an assumption or not motivating your assumptions will give you less points.

Also, remember to **write in a clear, readable handwriting**.

Good luck with the exam!

Problem Number	Solved	Points
1		
2		
3		
4		
5		
6		
Sum		
Bonus Points		
Grade		

Please set a X in the Solved column for each problem you have worked on.

Problem 1: Short Questions (10p)

In this problem you can give short answers without motivation, unless stated differently.

a. How does the CoDel protocol determine when to drop packets? Motivate your answer in one sentence.
   A: CoDel drops packets when the queue delay exceeds a pre-defined threshold, ensuring that the network does not accumulate excessive queueing delay.

b. Random variable X has a probability mass function of {0.25, 0.25, 0.25, 0.25}, and random variable Y has a probability mass function of {0.4, 0.2, 0.2, 0.2}. Which random variable has a higher entropy value?
   A: Thus, X has a higher entropy (2 bits) than Y (1.922 bits).

c. In a Binomial distribution, what happens to the distribution as $n \to \infty$ and $p \to 0$ while keeping np constant?
   A: The Binomial distribution approaches a Poisson distribution with parameter $\lambda = np$.

d. Your friend designs a source coding scheme where the entropy of the source is 1.95 bits, but the mean code length is 1.8 bits. Is this plausible? Motivate your answer in one sentence.
   A: No, it is not plausible because the mean code length cannot be smaller than the entropy due to Shannon's source coding theorem.

e. Mention one of the main components of information centric networking.
   A: Name resolution/routing, security, naming scheme for objects, forwarding and transporting.

f. Mention one advantage and one disadvantage of using active measurements in the network.
   A: Advantage: Active measurements provide precise performance evaluation.
   Disadvantage: They introduce additional traffic overhead.

g. Can Kubernetes automatically handle the internal communication between different pods within a cluster using its networking model?
   A: Yes, Pods are connected via an overlay network enabling IP-per-Pod networking and direct pod-to-pod communication (see slide 42 in the PDF of Simone's presentation).

h. What happens to the sample mean as the sample size increases, according to the Central Limit Theorem?
   A: As the sample size increases, the sample mean becomes more normally distributed, even if the original population is not normally distributed.

i. Mention three advantages and one disadvantage of using QUIC instead of TCP.
   A: Advantages: faster connection establishment, built-in encryption, multiplexing without head-of-line blocking.
   Disadvantage: overhead and complexity.

j. What distinguishes Discrete Event Simulation from Monte Carlo Simulation?

A: Discrete Event Simulation models the behavior of a system by representing events that occur at specific points in time, while Monte Carlo Simulation relies on random sampling to simulate a process.

Problem 2: Gaussian Channel Capacity (4p)

Consider a scenario with nodes A and B that communicate with node C using a Gaussian channel with a bandwidth of 10 MHz. Answer the following questions:

a. With only node A transmitting, what is the channel capacity for node A if its SNR is 10 dB? (1p)
b. With only node B transmitting, what should be the SNR, in decibels, to achieve a channel capacity of 50 Mbps? (1p)
c. Now, consider that nodes A and B maintain their transmit power and transmit at the same time. If we treat the signal from each node as interference for the other node (treating interference as noise), calculate the resulting information capacity for both nodes A and B. (2p)

Solution:

a. For node A, $P_A/N = 10^{10/10} = 10$ in linear scale. So, the channel capacity for node A is $10 \times 10^6 \times \log_2(1 + 10) = 10 \times 10^6 * 3.459 = 34.59$ Mbps. **(1p here)**
b. To achieve a capacity of 50 Mbps on node B, we set $50 \times 10^6 = 10 \times 10^6 \times \log_2\left(1 + \frac{P_B}{N}\right)$, such that $\log_2\left(1 + \frac{P}{N}\right) = 5$, and $P_B/N = 31$. This corresponds to a decibel value of $10 \times \log_{10}(31) = 14.91$ dB. **(1p here).**
c. For node A, consider signal from node B as noise. Hence, the information capacity is: $I_A = \log_2\left(1 + \frac{P_A}{N+P_B}\right)$, where $P_A = 10N$ and $P_B = 31N$. Hence, $I_A = \log_2\left(1 + \frac{10}{32}\right) = 0.3562$ bits/s. **(1p here).**
   For node B, consider signal from node A as noise. Hence, the information capacity is $I_B = \log_2\left(1 + \frac{P_B}{N+P_A}\right) = \log_2\left(1 + \frac{31}{11}\right) = 1.805$ bits/s. Note that there is almost no information from node A. **(1p here).**

Problem 3: Queueing Systems (6p)

Consider a network to which packets arrive as a Poisson process at a rate of 280 packets/s such that the time taken to service a packet is exponentially distributed. Suppose that the mean packet length is 500 bytes and that the link capacity is 1.4 Mbps. This network can accommodate at most three hosts, but it is able to improve fifty percent of its link capacity when there are more than two hosts in the system.

Based on this, answer the following questions:

a) Classify this queueing system according to Kendall's notation and draw the state transition diagram. (2p)
b) What are the service rates for this network at each state? (1p)
c) What is the probability that the system is empty? (1p)
d) What is the probability that the network needs to block hosts? (1p)
e) Calculate the expected number of hosts in the system. (2p)

Solution:

Notice that $\lambda = 280$ packets/s, $L = 500 * 8 = 4000$ bits/packet, and $R = 1.4 * 10^6$ bits/s.

a. This system is an M/M/1/3 queue with state-dependent service rates:
M: Poisson arrival process, M: Exponential service times, 1: Single server (the network), 3: Maximum capacity of 3 hosts. **(0.25p each, 1p in total).**

b. The packet length is 500 bytes = 4 000 bits, so the basic link service rate of $\mu = \frac{1.4*10^6 \text{ bits/s}}{4000 \text{ packets/s}}$ = 350 packets/s. Hence, $\mu = 350$ packets/s. With a 50% capacity increase when there are more than 2 hosts:
   - $\mu_1 = 350$ packets/s (states 0, 1, 2).
   - $\mu_2 = 350$ packets/s (states 0, 1, 2).
   - $\mu_3 = 350 \times 1.5 = 525$ packets/s (state 3, 50% increased capacity). **(0.33p each, 1p in total)**.

c. For a birth-death process with state-dependent rates, the steady-state probabilities are:
$P_1 = (\lambda/\mu_1)P_0$, $P_2 = (\lambda/\mu_1)(\lambda/\mu_2)P_0 = (\lambda/\mu_1)^2 P_0$ (since $\mu_1 = \mu_2$),
$P_3 = (\lambda/\mu_1)^2(\lambda/\mu_3)P_0 = (\lambda^2/{\mu_1}^2\mu_3)P_0$.
With $\lambda = 280, \mu_1 = \mu_2 = 350$, and $\mu_3 = 525$:
$P_1 = (280/350)P_0 = 0.8P_0, P_2 = (280/350)^2 P_0 = 0.64P_0$,
$P_3 = (280/350)^2(280/525)P_0 = 0.64 \times 0.533P_0 = 0.341P_0$.
Using the normalization condition ($P_0 + P_1 + P_2 + P_3 = 1$):
$P_0 + 0.8P_0 + 0.64P_0 + 0.341P_0 = 1$,
$2.781P_0 = 1, P_0 = 0.36$. **(1p here)**
The network needs to block hosts when it's at maximum capacity (state 3). The probability of blocking is: $P_3 = (\lambda/\mu_1)^2(\lambda/\mu_3)P_0 = (\lambda^2/{\mu_1}^2\mu_3)P_0$. Thus, $P_3 = 0.123$ or 12.3%. **(1p here).**

d. The expected number of hosts (L) is calculated using:
$L = \Sigma_{i=0}^{3}(i \times P_i)$, where: i is the number of hosts in state i, P_i is the probability of being in state i.
From our previous calculations: $P_0 = 0.36, P_1 = 0.288, P_2 = 0.23, P_3 = 0.123$. **(0.5p here)**.
Calculating L:

$$L = 0 \times P_0 + 1 \times P_1 + 2 \times P_2 + 3 \times P_3,$$

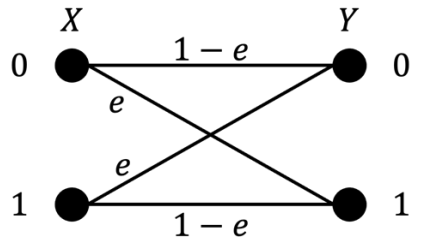
$$L = 0 \times 0.36 + 1 \times 0.288 + 2 \times 0.23 + 3 \times 0.123,$$

$L = 0 + 0.288 + 0.46 + 0.369, L = 1.117.$

Therefore, the expected number of hosts in the system is approximately 1.117 hosts. **(0.5p here)** This means that, on average, there are about 1.12 hosts in the network at any given time, which is between 1 and 2 hosts. This is consistent with the state probabilities we calculated, where the system spends most of its time in states 1 and 2, with less time in states 0 and 3.

Problem 4: Mutual Information and Channel Capacity (6p)

Consider a noisy channel that receives an input X that can be either 0 or 1 with equal probability $p = 0.5$. Due to noise the output will be corrupted with a probability $e = 0.02$. This channel is denoted as binary symmetric channel (BSC) and represented in the figure below.



Answer the following questions:

a) Calculate the entropy of the output Y. (1p)
b) Calculate the conditional entropy of the output Y given the input X. (1p).
c) Calculate the mutual information for this channel. (1p)
d) Prove that the capacity of the channel is $C = 1 - H(e)$ and compare its value to the mutual information. (3p)

Solution:

a) We calculate $H(Y)$ as the entropy of the output. For a BSC with $p(X = 0) = p(X = 1) = 0.5$ and error probability e:

$$p(Y = 0) = p(X = 0) \cdot p(Y = 0|X = 0) + p(X = 1) \cdot p(Y = 0|X = 1),$$
$$p(Y = 0) = 0.5 \times (1 - e) + 0.5 \times e,$$
$$p(Y = 0) = 0.5 \times 0.98 + 0.5 \times 0.02 = 0.49 + 0.01 = 0.5.$$ **(0.25p here)**

Similarly, p(Y=1) = 0.5. **(0.25p here)**

Therefore,

$$H(Y) = -[p(Y = 0) \cdot \log p(Y = 0) + p(Y = 1) \cdot \log p(Y = 1)],$$

$H(Y) = -[0.5 \cdot log(0.5) + 0.5 \cdot log(0.5)] = 1$ bit. **(0.5p here)**.

b) We obtain the conditional entropy $H(Y|X)$ as:

$$H(Y|X) = -\sum_x \sum_y p(x, y) \log p(y|x),$$

$$= -p(X = 0) \cdot [p(Y = 0|X = 0) \log p(Y = 0|X = 0) + p(Y = 1|X = 0) \log p(Y = 1|X = 0)]$$
$$-p(X = 1) \cdot [p(Y = 0|X = 1) \log p(Y = 0|X = 1) + p(Y = 1|X = 1) \log p(Y = 1|X = 1)],$$
$$H(Y|X) = -0.5 \cdot [(1 - e) \log(1 - e) + e \cdot \log(e)] - 0.5 \cdot [e \cdot \log(e) + (1 - e) \log(1 - e)],$$
$$H(Y|X) = -[0.5(1 - e) \log(1 - e) + 0.5e \cdot \log(e) + 0.5e \cdot \log(e) + 0.5(1 - e) \log(1 - e)],$$
$$H(Y|X) = -[(1 - e)log(1 - e) + e \cdot log(e)] = -[(0.98)log(0.98) + (0.02)log(0.02)],$$
$$H(Y|X) = -[0.98 \times (-0.02922) + 0.02 \times (-5.64386)] = -[-0.02864 - 0.11288],$$

$H(Y|X) = 0.14152$ bits **(1p here)**.

c) Now, we can calculate the mutual information $I(X; Y)$ as: $I(X; Y) = H(Y) - H(Y|X)$.
$I(X; Y) = H(Y) - H(Y|X) = 1 - 0.14152 = 0.85848$ bits. **(1p here)**.

d) The capacity of this channel with error probability e is given by: $C = 1 - H(e)$, where $H(e)$ is the binary entropy function: $H(e) = -[e \cdot \log(e) + (1 - e) \cdot \log(1 - e)]$. To prove the capacity, we do the following.
The mutual information can be written as: $I(X; Y) = H(Y) - H(Y|X)$
For BSC, $H(Y|X) = H(e)$ regardless of the input distribution, because the error probability is fixed at e: $H(Y|X) = -[(1 - e)log_2(1 - e) + e \cdot log_2(e)] = H(e)$.

To maximize $I(X;Y)$, we need to maximize $H(Y)$ given that $H(Y|X)$ is constant.
Let $p = P(X = 0)$ and $1 - p = P(X = 1)$. Then:

$$P(Y = 0) = p(1 - e) + (1 - p)e, P(Y = 1) = pe + (1 - p)(1 - e).$$

$$H(Y) = -P(Y = 0)logP(Y = 0) - P(Y = 1)logP(Y = 1).$$

The maximum value of $H(Y)$ is 1 bit, which occurs when $P(Y = 0) = P(Y = 1) = 0.5$.
This happens when p = 0.5, i.e., when the input distribution is uniform.
Therefore:
$C = max_p\, I(X;Y) = max_p\, [H(Y) - H(e)] = 1 - H(e)$. **(2p here)**

Now, using $e = 0.02$:

$$C = 1 - H(0.02) = 1 - (-[0.02 \cdot log(0.02) + 0.98 \cdot log(0.98)]),$$

$C = 0.85848$ bits.
The mutual information equals the capacity of the channel. This is expected because the capacity is achieved when input symbols are equally likely (p = 0.5), which is exactly our scenario. **(1p here)**.

Problem 5: Markov Chains (5p)

Consider a network node with four states that represent the status of its connection: disconnected, idle, active, and reconnecting. The node transitions between these states according to the following probabilities:

- Disconnected (D): The node cannot communicate with other nodes.
  - It remains disconnected with a probability of 0.6.
  - It transitions to the idle state with a probability of 0.2.
  - It transitions to the reconnecting state with a probability of 0.2.
- Idle (I): The node is connected but not actively transmitting data.
  - It stays idle with a probability of 0.5.
  - It transitions to the active state with a probability of 0.3.
  - It transitions to the disconnected state with a probability of 0.1.
  - It transitions to the reconnecting state with a probability of 0.1.
- Active (A): The node is actively transmitting data.
  - It remains active with a probability of 0.4.
  - It transitions to the idle state with a probability of 0.2.
  - It transitions to the disconnected state with a probability of 0.3.
  - It transitions to the reconnecting state with a probability of 0.1.
- Reconnecting (R): The node is trying to reconnect to the network.
  - It stays in the reconnecting state with a probability of 0.5.
  - It transitions to the idle state with a probability of 0.3.
  - It transitions to the active state with a probability of 0.1.
  - It transitions to the disconnected state with a probability of 0.1.

Now, answer the following questions:

a) Derive the state transition matrix. (1p)

b) Does the Markov Chain exhibit any periodic behavior? Justify your answer. If periodic behavior is present, suggest how to modify the system to avoid it. (1p)

c) Given that the node starts in the idle state, compute the probabilities that the node is in each state after one and two seconds. (2p)

d) Write a pseudocode to compute the stationary distribution of the Markov Chain. (1p)

Solution:

a) The state transition matrix is: $P = \begin{pmatrix} 0.6 & 0.2 & 0 & 0.2 \\ 0.1 & 0.5 & 0.3 & 0.1 \\ 0.3 & 0.2 & 0.4 & 0.1 \\ 0.1 & 0.3 & 0.1 & 0.5 \end{pmatrix}$ **(1p here)**.

b) All states have transitions to all other states (either directly or indirectly). Since there is no fixed step count to return to a particular state, and transitions occur in different ways with different probabilities, the Markov Chain is aperiodic. **(1p here)**.

c) Consider that $\pi^0 = [0, 1, 0, 0]$. For one second, we compute $\pi^1 = \pi^0 P = [0.1, 0.5, 0.3, 0.1]$ **(1p here)** and $\pi^2 = \pi^1 P = [0.23, 0.34, 0.28, 0.15]$ **(1p here)**.

d) See the image below:

```python
# For the stationary distribution, do this for n times
n_station = 100
staPiP =  pi
for idx in range(n_station):
    staPiP = staPiP @ P

print("Stationary distribution:", staPiP)
```

Problem 6: ALOHA and Slotted ALOHA (6p)

A student sets up a simple Slotted ALOHA-style MAC protocol for a small wireless network in their room. The network has two backlogged nodes, labeled X and Y. The student assigns a transmission probability of 2q for node X and q for node Y. Each packet occupies one time slot, and all transmissions start at the beginning of a slot. Now, answer the following questions:

a. Determine the value of q that maximizes the network's utilization, and calculate the maximum utilization. Justify your answer. (3p)
b. How would your answer to part a. change if the network used ALOHA instead of Slotted ALOHA? Explain your reasoning, and calculate the value of q that maximizes the network's utilization and the maximum utilization. (3p)

Solution:

a) In a Slotted ALOHA system with two nodes X and Y, where transmission probabilities are 2q and q respectively, the utilization U can be calculated as:
U = P(exactly one node transmits) U = P(X transmits and Y doesn't) + P(Y transmits and X doesn't)
$U = 2q(1-q) + q(1-2q)\, U = 2q - 2q^2 + q - 2q^2\, U = 3q - 4q^2$. **(1p here)**
To find the value of q that maximizes utilization, I differentiate U with respect to q and set it to zero:
$dU/dq = 3 - 8q = 0\, 8q = 3, q = 3/8 = 0.375$. **(1p here)**
To verify this is a maximum, we check the second derivative:
$d^2U/dq^2 = -8 < 0$, confirming this is a maximum.
The maximum utilization is:

$$U = 3\left(\frac{3}{8}\right) - 4\left(\frac{3}{8}\right)^2 = \frac{9}{8} - 4\left(\frac{9}{64}\right) = \frac{9}{8} - \frac{36}{64} = \frac{72}{64} - \frac{36}{64},$$
$$U = 36/64 = 9/16 = 0.5625.$$

Therefore, q = 3/8 maximizes the network utilization at 0.5625 or 56.25%. **(1p here)**

b) In pure ALOHA, we can adapt the analysis from part (a) but need to account for the differences between slotted and unslotted protocols. For unslotted ALOHA with two nodes X and Y with transmission probabilities 2q and q respectively.
The utilization equation remains similar to part (a), but the probability of collision changes: U = P(X transmits successfully) + P(Y transmits successfully), which is:
$U = 2q(1-q)^2 + q(1-2q)^2$.
This is because in unslotted ALOHA, for a transmission to be successful, no other node should transmit during the vulnerable period, which includes both the current and adjacent time periods. **(1p here)**
To find the maximum utilization, we differentiate U with respect to q and set it to zero:

$$U = 2q(1-q)^2 + q(1-2q)^2 \rightarrow U = 2q(1-2q+q^2) + q(1-4q+4q^2),$$
$$U = 2q - 4q^2 + 2q^3 + q - 4q^2 + 4q^3 \rightarrow U = 3q - 8q^2 + 6q^3,$$
$$dU/dq = 3 - 16q + 18q^2 = 0.$$

Solving this quadratic equation: $18q^2 - 16q + 3 = 0$.
Using the quadratic formula:

$$q = [16 \pm \sqrt{(256-216)}]/36\; q = [16 \pm \sqrt{40}]/36\; q = [16 \pm 6.32]/36.$$

This gives $q \approx 0.619$ or $q \approx 0.270$. The second derivative test confirms $q \approx 0.270$ gives a maximum. **(1p here)**.
Substituting q = 0.270 into the utilization equation:

$$U = 3(0.270) - 8(0.270)^2 + 6(0.270)^3 \rightarrow U = 0.810 - 8(0.0729) + 6(0.0197),$$
$$U = 0.810 - 0.5832 + 0.1182,$$
$$U = 0.345 \text{ or } 34.5\% \textbf{ (1p here)}.$$

Therefore, in unslotted ALOHA, $q \approx 0.270$ maximizes the network utilization at approximately 34.5%, which is lower than the 56.25% achievable in Slotted ALOHA.